20L-0961

Natural Language Processing Homework 1

Term Frequency (TF) = No of times the term appears in document / total number of terms in document

Inverse Document Frequency (IDF) = $\log\left(\frac{N}{DF}\right) = \log\left(\frac{\text{Number of Documents}}{\text{Number of Documents that contains the word}}\right)$

TF-IDF Table for Question

DocID	Text	TF	IDF	TF-IDF
	itsy	1/4	$\lg(16/4)$	0.125
	bitsy	1/4	$\lg(16/4)$	0.15
	spider	1/4	$\lg(16/6)$	0.11
1	water	0/4	$\lg(16/2)$	0
	sprout	0/4	$\lg(16/4)$	0
	went	0/4	$\lg(16/4)$	0
	up	0/4	$\lg(16/4)$	0
	itsy	0/6	$\lg(16/4)$	0
	bitsy	0/6	$\lg(16/4)$	0
	spider	1/6	$\lg(16/6)$	0.07
4	water	0/6	$\lg(16/2)$	0
	sprout	0/6	$\lg(16/4)$	0
	went	0/6	$\lg(16/4)$	0
	up	0/6	$\lg(16/16)$	0

	itsy	1/6	lg(16/4)	0
	bitsy	1/6	lg(16/4)	0
	spider	1/6	lg(16/6)	0.07
7	water	0/6	lg(16/2)	0
	spout	0/6	lg(16/4)	0
	went	0/6	lg(16/4)	0
	up	0/6	lg(16/6)	0

Cosine similarity of $D_1, D_4, 1 \ldots$

$$\text{cosine}(A,B) = \frac{A \cdot B}{|A| \cdot |B|}$$

$$\text{cosine}(D_1, D_4) = \frac{\{1,1,1,0,0,0,0\} \cdot \{0,0,1,0,0,0,0\}}{\sqrt{1^2+1^2+1^2+0^2+0^2+0^2+0^2}\ \sqrt{0^2+0^2+1^2+0^2+0^2+0^2+0^2}}$$

$$= \frac{0+0+1+0+0+0+0}{\sqrt{3} \cdot \sqrt{1}} = \frac{1}{\sqrt{3}} = 0.57$$

$$\text{cosine}(D_1, D_7) = \frac{\{1,1,1,0,0,0,0\} \cdot \{1,1,1,0,0,0,0\}}{\sqrt{1^2+1^2+1^2+0^2+0^2+0^2+0^2}\ \sqrt{1^2+1^2+1^2+0^2+0^2+0^2+0^2}}$$

$$= \frac{1+1+1}{\sqrt{3}\sqrt{3}} = \frac{3}{\sqrt{3}} = 3/3 = 1$$

$$\text{cosine}(D_4, D_7) = \frac{\{0,0,1,0,0,0,0\} \cdot \{1,1,1,0,0,0,0\}}{\sqrt{0^2+0^2+1^2+0^2+0^2+0^2+0^2}\ \sqrt{1^2+1^2+1^2+0^2+0^2+0^2+0^2}}$$

$$= \frac{0+0+1+0+0+0+0}{\sqrt{1}\ \sqrt{3}} = \frac{1}{\sqrt{3}} = 0.57$$